

# NEXUS — AI-Powered Criminal Network Analysis

## Comprehensive Project Architecture, PPT Blueprint & Demo Proof Guide

**Project Tagline:** “Connect the clues. Reveal the network.”  
**Core Purpose:** Transform fragmented police records, CDR logs, banking transactions, national identities, and CCTV sightings into a unified, explainable 3D investigation network.  
**Live Web Portal:** <https://itzhrni.github.io/nexus-investigation-ai/> (Zero setup, publicly accessible).

### 1. The Core Philosophy: “PPT Explains; Demo Proves”

A common pitfall in hackathons and high-stakes evaluations is duplicating content between the presentation and the demo video. When the video repeats slides and objectives, valuable time is lost. NEXUS strictly enforces the **Complementary Division Principle**:

| Dimension      | PPT Presentation (“The Explanation”)                                                                                                                                                                                                                                    | Demo Video (“The Proof”)                                                                                                                                                                                                                                            |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Role           | Explains the <b>Idea, Problem, Architecture &amp; Impact</b>                                                                                                                                                                                                            | Proves the <b>Execution, Working Prototype &amp; Performance</b>                                                                                                                                                                                                    |
| What it covers | <ul style="list-style-type: none"><li>• Problem landscape (silos, cross-state gaps)</li><li>• Mathematical &amp; AI methodology (\$D_5\$, Louvain)</li><li>• Legal frameworks (Aadhaar Act 2016, PMLA)</li><li>• Scalability, cost &amp; deployment viability</li></ul> | <ul style="list-style-type: none"><li>• Real UI interaction on live browser portal</li><li>• Any-Clue Search live resolving a phone / Aadhaar</li><li>• 3D Graph expanding from 1 to 5 hops</li><li>• Instant detection of mule collisions &amp; fake IDs</li></ul> |
| Tone & Format  | Strategic, analytical, structured, authoritative slides                                                                                                                                                                                                                 | High-tempo, compelling, 100% screen-captured proof                                                                                                                                                                                                                  |
| What to AVOID  | DO NOT show 30 screenshots of code or menus                                                                                                                                                                                                                             | DO NOT spend 1 second showing slides or reading goals                                                                                                                                                                                                               |

### 2. Complete Technical Architecture & Newly Added Modules

Below is the complete inventory of NEXUS modules, explicitly incorporating all capabilities recently implemented:

#### A. Any-Clue Search & Bharat-Aware Entity Resolution

- **Heterogeneous Clue Entry:** Investigators can initiate queries via Person name, Phone, SIM ICCID, Device IMEI, Vehicle Number Plate, Bank Account Number, FIR number, or national Aadhaar ID.
- **Bharat-Aware Multilingual NLP:** Transliteration, fuzzy phonetic Soundex/Metaphone matching across Indian languages (Hindi, Tamil, Marathi, Telugu, Bengali). Handles spelling variances (e.g., *Arun Kumar* vs. ■■■■ ■■■■■■).

#### B. Aadhaar Forensics & Mule Anti-Fraud Engine (NEW IMPLEMENTATION)

- **Verhoeff Algorithm (\$D\_5\$ Dihedral Group):** Validates check-digits via dihedral permutations, catching **100% of single-digit errors and adjacent transposition errors**. Random fabricated numbers immediately fail checksum.
- **Aadhaar Act 2016 Privacy Compliance:** Zero plaintext storage. All records masked as XXXX-XXXX-1234 with deterministic SHA-256 salted hashing for secure relational indexing.

- **1-to-Many Collision & Mule Fan-out:** Detects when multiple distinct personas claim the same credential. Automatically calculates mule SIM and bank account fan-out across the criminal syndicate.

### C. Dynamic 3D Focal Investigation Graph & Topological Analytics

- **PostgreSQL Recursive CTE:** Fast multi-hop traversals (1 to 5 hops) executed via recursive SQL Common Table Expressions directly in PostgreSQL, eliminating external graph database license dependencies.
- **Louvain Community Detection:** NetworkX-powered modularity optimization partitioning complex syndicates into sub-cluster cells.
- **Betweenness Centrality Bridge Detection:** Pinpoints crucial liaison operatives and hawala money couriers connecting otherwise isolated criminal syndicates.

### D. Real Interactive Indian Geospatial Map (Leaflet GIS)

- **Indian Territory & Police Station Mapping:** Geocoded markers for Delhi, Mumbai, Bengaluru, Chennai, Jaipur, Lucknow, Kolkata, and Hyderabad. Displays jurisdictional limits and police station boundaries.
- **Bi-Directional Graph ↔ Map Interaction:** Clicking a node on the graph zooms into its state/district footprint on the map; selecting a map pin highlights connected entities in the 3D graph.

### E. Investigation Analytics & Automated Pattern Detection

- **Deterministic Heuristic Detection:** Automatically flags *SIM Switching*, *Circular Financial Loops*, *Multi-Vehicle Location Convergence*, and *Cross-Jurisdiction Boundary Hopping*.
- **Temporal 'What Changed?' Engine:** Quantifies surge anomalies before and after critical incident timestamps (transaction volume surges, call frequency spikes).

### F. Court-Admissible Dossier Export, Auth & Audit Logging

- **1-Click Executive Report Dossier:** Client-side jsPDF + JSZip exporter compiling full 6-W evidence briefs, timeline CSVs, JSON graph snapshots, and PNG visualizations into a court-ready dossier.
- **Enterprise Security:** Signed HMAC-SHA256 JWT authentication, RBAC (Investigator vs Admin), and immutable PostgreSQL audit\_logs tracking officer queries for Section 65B compliance.

### 3. Slide-by-Slide PPT Presentation Checklist

Designed specifically for national hackathons (e.g. SIH 5-Slide Template or 5-Minute Pitch Decks). Use this checklist to ensure your presentation explains the *why* and *how*, setting up the demo video to prove it works.

#### SLIDE 1: SOLUTION OVERVIEW & PROBLEM STATEMENT

**Key Objective:** Frame the urgent national security problem and introduce NEXUS as the unified answer.

**Essential Items to Include:**

- **Problem:** Fragmented law enforcement data silos (CCTNS, CDR logs, bank RTGS, toll FASTag, FIRs in regional languages).
- **Core Bottleneck:** Inter-state jurisdictional boundaries cause criminals to rotate SIMs, hop vehicles, and exploit mule accounts.
- **NEXUS Solution:** An AI-powered, Any-Clue Investigation platform transforming isolated records into a unified 3D graph.
- **System Architecture Diagram:** High-level flow showing Data Ingestion → AI Entity Resolution → Recursive CTE Graph → 6-W Evidence Output.

#### SLIDE 2: TECHNICAL APPROACH & MATHEMATICAL INNOVATION

**Key Objective:** Convince technical judges of your engineering rigor and algorithmic depth.

**Essential Items to Include:**

- **PostgreSQL Recursive CTE Graph:** Graph traversal natively in SQL without expensive proprietary Neo4j licenses.
- **Verhoeff \$D\_5\$ Dihedral Group Checksum:** Mathematical check-digit algorithm (\$d, p, inv\$ tables) catching 100% of single-digit and adjacent transposition forgery.
- **NetworkX Louvain & Betweenness Centrality:** Graph analytics discovering hidden criminal cells and bridge liaisons.
- **Bharat-Aware NLP & Transliteration:** Multi-language phonetic normalization resolving aliases across regional scripts.

#### SLIDE 3: FEASIBILITY, VIABILITY & LEGAL COMPLIANCE

**Key Objective:** Prove that the platform can legally and operationally deploy in Indian law enforcement.

**Essential Items to Include:**

- **Aadhaar Act 2016 Compliance:** Masked storage (XXXX-XXXX-1234) and salted SHA-256 hashes—zero plaintext leaks.
- **Explainable 6-W Evidence (No Black-Box AI):** Produces transparent *Who, What, When, Where, Why, and Confidence* justifications acceptable in court.
- **Indian Evidence Act Section 65B Readiness:** Immutable cryptographic audit logs verifying chain of custody.
- **Deployment Flexibility:** Dual-engine support—enterprise PostgreSQL backend + instant zero-setup browser deployment.

#### SLIDE 4: LAW ENFORCEMENT IMPACT & MEASURABLE BENEFITS

**Key Objective:** Highlight tangible efficiency gains and crime deterrence value.

**Essential Items to Include:**

- **Latency Reduction:** Reduces manual entity correlation from **14–21 days to under 2 seconds**.
- **Inter-State Border Synergy:** Real-time GIS alerts across state police departments (e.g. Haryana → UP → Delhi).
- **Mule Account Disruption:** Unmasks synthetic identity rings and fan-out mule networks before funds leave the country.
- **Resource Optimization:** Automates tedious FIR/CDR cross-referencing so investigators focus on field action.

#### SLIDE 5: RESEARCH, REFERENCES & ROADMAP

**Key Objective:** Anchor the system in established academic research and official standards.

**Essential Items to Include:**

- **Algorithmic Citations:** J. Verhoeff (1969) Check-Digit Systems; Blondel et al. (2008) Louvain Fast Unfolding of Communities; U. Brandes (2001) Faster Algorithm for Betweenness Centrality.
- **Statutory & Government Standards:** UIDAI Aadhaar Architecture Specs; FIU-IND PMLA Typology Reports; CCTNS Interoperability Guidelines.
- **Future Roadmap:** Automated CCTNS live API integration, facial recognition OCR integration, and multi-tenant department federation.

### 4. 3-Minute Demo Video Execution Script (“The Proof”)

Remember: **PPT explains; Demo proves.** Do not show title slides or recite problem statements in the video. Jump directly into the live software on screen and let the features prove your capability in 180 seconds:

| Timestamp               | Visual Demonstration (On Screen)                                                                                                                                                                                                                                                                                                                                                                         | Audio Narration Cue (Spoken Voiceover)                                                                                                                                                                                                                          |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0:00 – 0:45<br>(45 sec) | <ul style="list-style-type: none"><li>Screen opens on <b>Any-Clue Search</b>.</li><li>Type suspect clue: XXXX-XXXX-9015 or phone +919811055443.</li><li>Hit <b>RESOLVE</b>. Show instant entity resolution matching <b>Karan Singh</b> with [COLLISION] badge.</li><li>Click <b>'Open in Graph'</b>.</li></ul>                                                                                           | <i>“Instead of isolated databases, NEXUS allows an officer to begin with any single clue. We input a suspected Aadhaar identifier—our engine instantly resolves candidate identities across state registries in milliseconds.”</i>                              |
| 0:45 – 1:30<br>(45 sec) | <ul style="list-style-type: none"><li>3D Graph renders live. Rotate graph in 3D orbit mode.</li><li>Click on <b>Karan Singh (P005)</b>: Right panel opens.</li><li>Show <b>Aadhaar Forensics Card</b>: Point out amber <b>COLLISION DETECTED</b> alert showing ID shared with <b>Pooja Bhatnagar (P015)</b>.</li><li>Click <b>Vikram Patel (P004)</b>: Show red <b>CHECKSUM INVALID</b> badge.</li></ul> | <i>“On the 3D network, NEXUS reveals structural relationships. Looking at Karan Singh, our Verhoeff anti-fraud engine flags an active collision—the same Aadhaar is simultaneously claimed by an operative in West Bengal, exposing a synthetic mule ring.”</i> |
| 1:30 – 2:15<br>(45 sec) | <ul style="list-style-type: none"><li>Switch to <b>Map Workspace</b> via sidebar.</li><li>Show Leaflet India GIS Map with pins in Delhi, Haryana, and Rajasthan.</li><li>Click on a police station / vehicle sighting marker.</li><li>Switch to <b>Timeline Workspace</b>: show temporal 'What Changed?' baseline surge (■10K to ■4.5L).</li></ul>                                                       | <i>“Switching to the GIS workspace, inter-state movements across state police boundaries are dynamically correlated. The temporal engine highlights critical financial surges exactly when the suspect changed SIM cards.”</i>                                  |
| 2:15 – 3:00<br>(45 sec) | <ul style="list-style-type: none"><li>Open <b>Analytics Workspace</b>: show Louvain community clusters and betweenness centrality bridge nodes.</li><li>Navigate to <b>Report Center</b>.</li><li>Click <b>'Export Investigation Dossier'</b>.</li><li>Open downloaded PDF dossier showing court-admissible 6-W evidence &amp; Section 65B hash.</li></ul>                                               | <i>“Finally, the Analytics workspace isolates network bridge couriers, and in one click, exports a court-admissible Section 65B evidentiary dossier. NEXUS turns scattered clues into swift, explainable justice.”</i>                                          |

### 5. Live Demo Test Profiles Cheat Sheet

Keep this quick-reference table handy while recording the demo video or demonstrating live to judges:

| Entity ID / Name            | Input Query to Type           | Forensic Result on Screen                       | Key Investigative Takeaway                    |
|-----------------------------|-------------------------------|-------------------------------------------------|-----------------------------------------------|
| <b>P001</b><br>Aarav Sharma | XXXX-XXXX-0016 or P001        | <b>VERHOEFF VALID (D5)</b><br>100% Tamper Proof | Clean legitimate citizen / victim profile.    |
| <b>P005</b><br>Karan Singh  | XXXX-XXXX-9015 or Karan Singh | <b>COLLISION DETECTED</b><br>Shared with P015   | Mule ID theft; 2 suspects claiming 1 Aadhaar. |

|                             |                                  |                                                |                                                         |
|-----------------------------|----------------------------------|------------------------------------------------|---------------------------------------------------------|
| <b>P011</b><br>Meera Mehta  | XXXX-XXXX-0114 or<br>Meera Mehta | <b>COLLISION DETECTED</b><br>Shared with P031  | Inter-state mule ring linking<br>Rajasthan & Telangana. |
| <b>P004</b><br>Vikram Patel | XXXX-XXXX-4048 or<br>P004        | <b>CHECKSUM INVALID</b><br>D5 Algorithm Failed | Fabricated check digit; forged<br>physical card.        |
| <b>P010</b><br>Suresh Nair  | XXXX-XXXX-9999 or<br>P010        | <b>CHECKSUM INVALID</b><br>Tampered Digits     | Synthetic fraudulent profile; fake<br>documentation.    |